

04_UI.md

SpeechStudio User Interface Specification

Version: 1.0

Status: Approved

1. Purpose

This document defines the complete graphical user interface of SpeechStudio.

The UI is the only interface between the user and the Higgs Audio V3 TTS engine.

Users must never edit Python files, configuration files or prompt tokens during normal operation.

The interface must expose every officially supported Higgs Audio V3 capability.

2. Design Goals

The interface shall be

Modern

Clean

Fast

Responsive

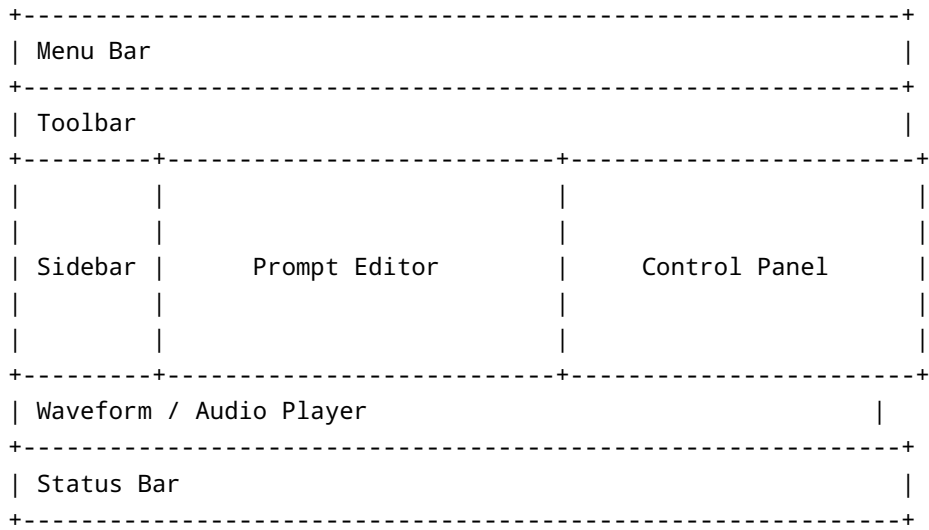
Dark Theme by default

Optimized for mouse and keyboard

Suitable for both beginners and advanced users

3. Main Window Layout

The application consists of five primary regions.



4. Menu Bar

File

New Project

Open Project

Save Project

Import Voice

Export Voice

Import Preset

Export Preset

Exit

Generation

Generate

Stop

Replay

Clear

Tools

Benchmark

Voice Library

History

Settings

Developer

View

Theme

Reset Layout

Fullscreen

Help

Documentation

About

5. Toolbar

Buttons

Generate

Stop

Replay

Open Output Folder

Voice Library

History

Settings

Benchmark

Developer Mode

Toolbar icons only.

Tooltip required.

6. Left Sidebar

Contains

Voice Library

History

Presets

Projects

Output Browser

Collapsible.

Resizable.

7. Center Panel

Large Prompt Editor.

Features

Plain text editing

Undo

Redo

Copy

Paste

Drag and Drop

Line numbers optional

Syntax highlighting in Developer Mode

Supports large prompts.

8. Right Control Panel

Scrollable.

Contains every Higgs configuration.

Sections

Voice

Emotion

Style

Prosody

Sound Effects

Generation

Reference Audio

Prompt Preview

9. Voice Section

Controls

Voice Profile selector

Import Voice

Export Voice

Create Voice

Delete Voice

Reference Audio

Reference Transcript

Voice Information

Display

Voice name

Reference duration

Sample rate

Channels

Status

10. Emotion Section

Contains every official Higgs emotion.

21 Toggle Buttons.

Exactly one selected in normal mode.

Supported

Elation

Amusement

Enthusiasm

Determination

Pride

Contentment

Affection

Relief

Contemplation

Confusion

Surprise

Awe

Longing

Arousal

Anger

Fear

Disgust

Bitterness

Sadness

Shame

Helplessness

Each button displays

Name

Tooltip

Example usage

Generated token

11. Style Section

Three buttons.

Singing

Shouting

Whispering

Exactly one active.

12. Prosody Section

Speed

Normal

Very Slow

Slow

Fast

Very Fast

Pitch

Normal

Low

High

Delivery

Normal

Expressive High

Expressive Low

Pause controls

Insert Pause

Insert Long Pause

Pause buttons insert inline tokens at cursor position.

13. Sound Effects

Official Higgs SFX only.

Cough

Laughter

Crying

Screaming

Burping

Humming

Sigh

Sniff

Sneeze

The UI automatically inserts

Token

Suggested onomatopoeia

Correct spacing

User never types SFX tokens manually.

14. Generation Section

Parameters

Temperature

Top P

Top K

Max New Tokens

Seed

Append Silence

Output Format

Auto Play

Normalize Audio

Each parameter contains

Current value

Default value

Tooltip

Reset button

15. Reference Audio

Reference WAV

Reference Transcript

Trim Reference

Normalize

Refresh

Display

Duration

Sample Rate

Channels

File Name

Validation Status

16. Prompt Preview

Read only.

Displays

Final prompt

Inserted Higgs tokens

Reference information

Warnings

Copy button

Developer Mode may allow editing.

17. Audio Player

Waveform

Play

Pause

Stop

Seek

Volume

Zoom

Playback Speed

Displays

Length

Sample Rate

Output Format

18. Status Bar

Displays

Model

CUDA

GPU Name

VRAM Usage

Generation Time

Generation Status

Current Voice

19. History Window

Each entry stores

Timestamp

Voice

Prompt

Generation Parameters

Output File

Duration

Replay

Reuse

Delete

Export

20. Preset Window

Stores

Voice

Emotion

Style

Prosody

Generation Parameters

Prompt Template

User can

Create

Rename

Duplicate

Delete

Import

Export

21. Settings Window

General

Generation

Paths

Audio

Performance

Developer

Appearance

No Python editing required.

22. Benchmark Window

Compare

Temperature

Top P

Top K

Emotion

Prosody

Voice

Displays

Generation Time

Output Duration

Realtime Factor

23. Developer Mode

Displays

Raw Prompt

Token List

GPU Memory

Generation Log

Prompt Validation

Timing

No additional functionality.

Only additional information.

24. Validation

Before generation

Prompt not empty

Voice valid

Reference valid

Generation parameters valid

Model loaded

CUDA available

Output path valid

Errors shown inside UI.

No terminal required.

25. Responsive Behaviour

Panels resizable.

Sidebar collapsible.

Right panel collapsible.

Layout automatically restored at startup.

26. Theme

Dark theme default.

Light theme optional.

Consistent spacing.

Rounded controls.

Native Windows behavior.

27. Keyboard Shortcuts

Ctrl N

New Project

Ctrl O

Open

Ctrl S

Save

Ctrl Enter

Generate

Esc

Stop

Space

Play Pause

Ctrl L

Voice Library

Ctrl H

History

Ctrl P

Prompt Preview

28. UI Principles

The interface must expose every officially supported Higgs Audio V3 feature.

The interface must not expose unsupported or experimental options.

Every visible control must correspond to either

an official Higgs prompt token

or

an officially supported generation parameter.

No placeholder controls.

No future functionality.

No hidden experimental features.

The UI is a graphical representation of the Higgs Audio V3 capabilities.